

## Training Configuration and Summary:

Dataset used: rssi\_odom\_dataset\_12apr25\_ajust.parquet

model\_type: GRU  
group\_sec: 80  
timesteps\_orig: 60  
batch\_size: 256  
min\_units: 16  
max\_units: 1024  
units\_step: 16  
dropout\_min: 0.2  
dropout\_max: 0.3  
dropout\_step: 0.1  
activation\_functions: ['tanh', 'relu', 'elu']  
optimizers: ['adam', 'rmsprop', 'Nadam']  
min\_layers: 1  
max\_layers: 2  
max\_epochs: 150  
hyperband\_iterations: 3  
group\_msec: 80  
samples\_per\_sec: 12  
forecast\_horizon\_sec: 4  
forecast\_steps: 48  
timesteps: 60

### GRU Multi-Step Training Report

Report Generated: 2025-05-21 08:15:47

Training Start: 2025-05-20 23-10-21

Training End: 2025-05-21 01-48-28

Training Duration: 158.12 minutes

Predictions Start: 2025-05-21 01-48-28

Predictions End: 2025-05-21 01-50-36

Predictions Duration: 2.13 minutes

#### Best Hyperparameters:

Number of Layers: 2

Activation: elu

Optimizer: adam

Dropout Rate: 0.20

MAE: 1.9750

MSE: 6.0364

RMSE: 2.4569

R2 Score: 0.6370

#### Multi-Step Forecast:

Forecast Steps: 48

MAE: 0.0216

MSE: 0.0007

RMSE: 0.0268

R2 Score: 0.9223

Dataset size before down-sample: 85,269 rows  
Dataset size after down-sample: 12,829 rows

First 5 timestamps BEFORE:

1744397085760  
1744397085770  
1744397085789  
1744397085801  
1744397085817

...

Last 5 timestamps BEFORE:

1744464971366  
1744464971377  
1744464971390  
1744464971402  
1744464971414

First 5 timestamps AFTER:

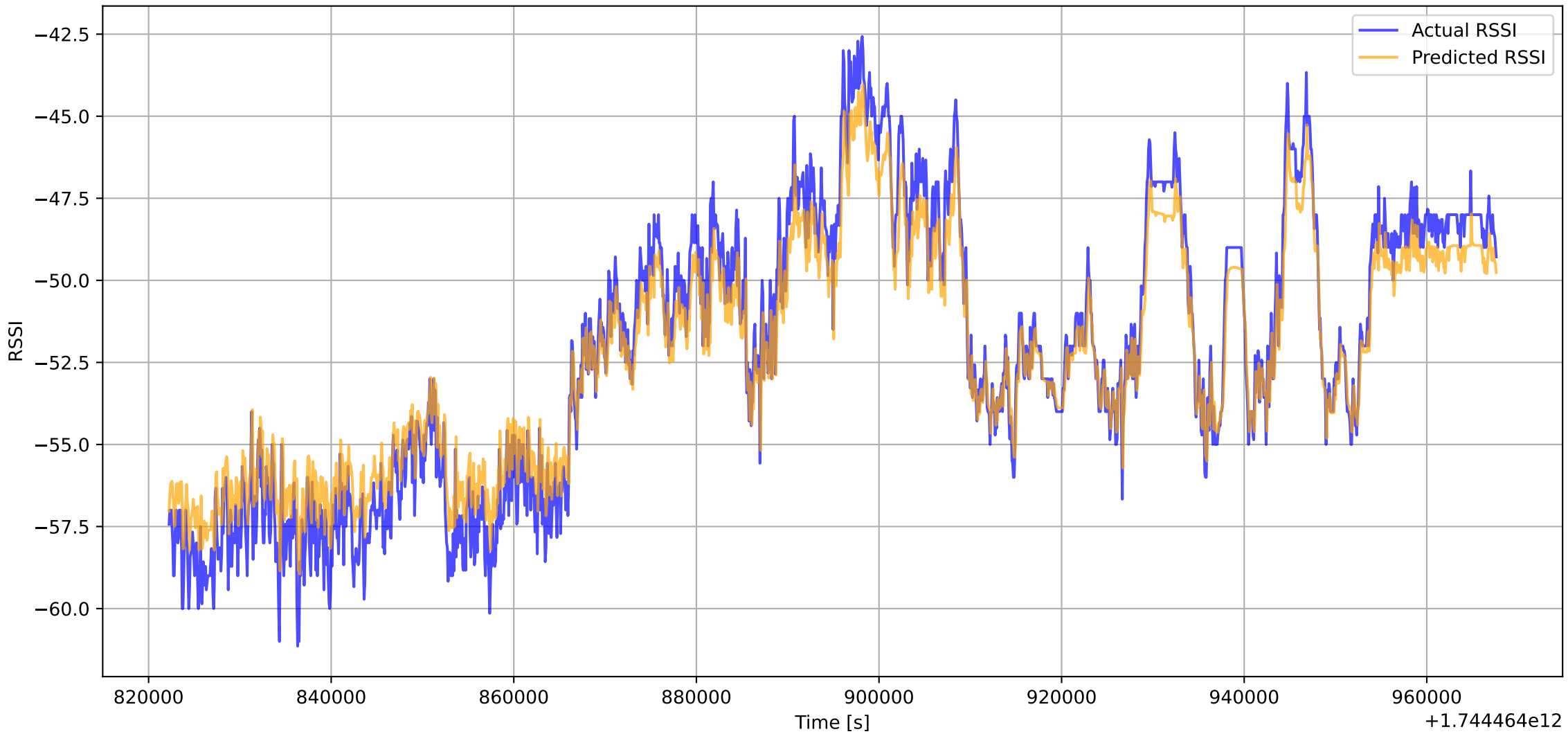
1744397085760  
1744397085841  
1744397086014  
1744397086103  
1744397086172

...

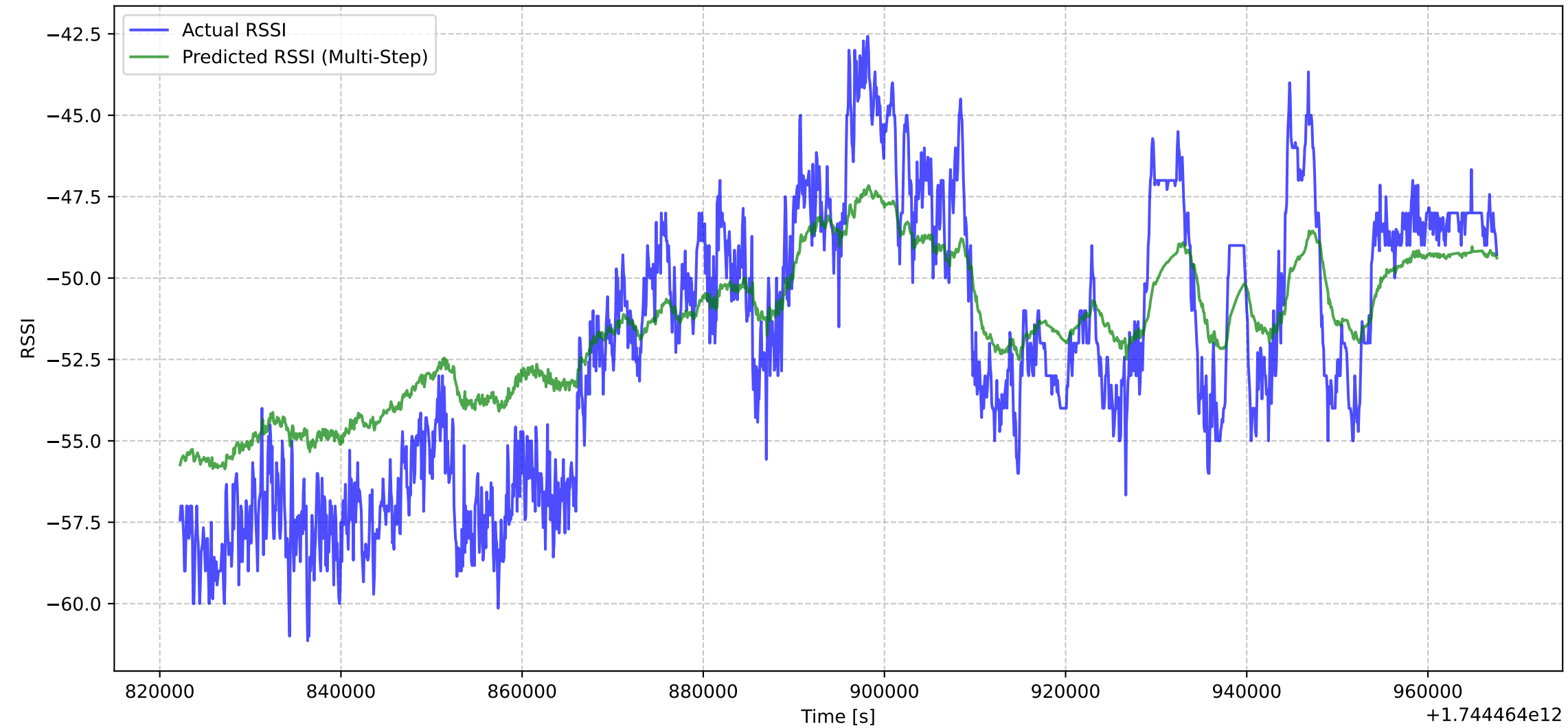
Last 5 timestamps AFTER:

1744464971041  
1744464971124  
1744464971209  
1744464971281  
1744464971366

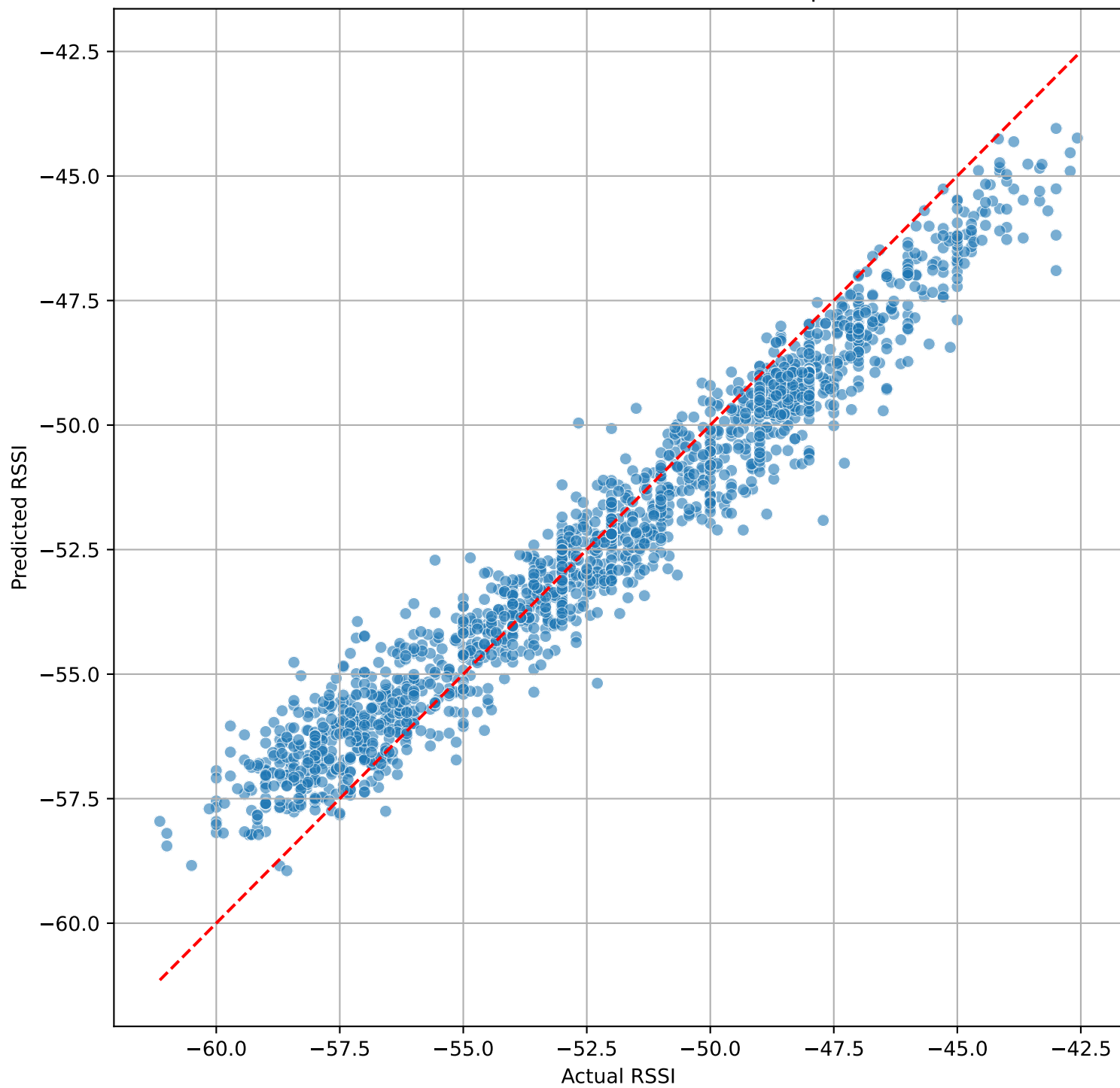
Actual vs Predicted RSSI (First Step)



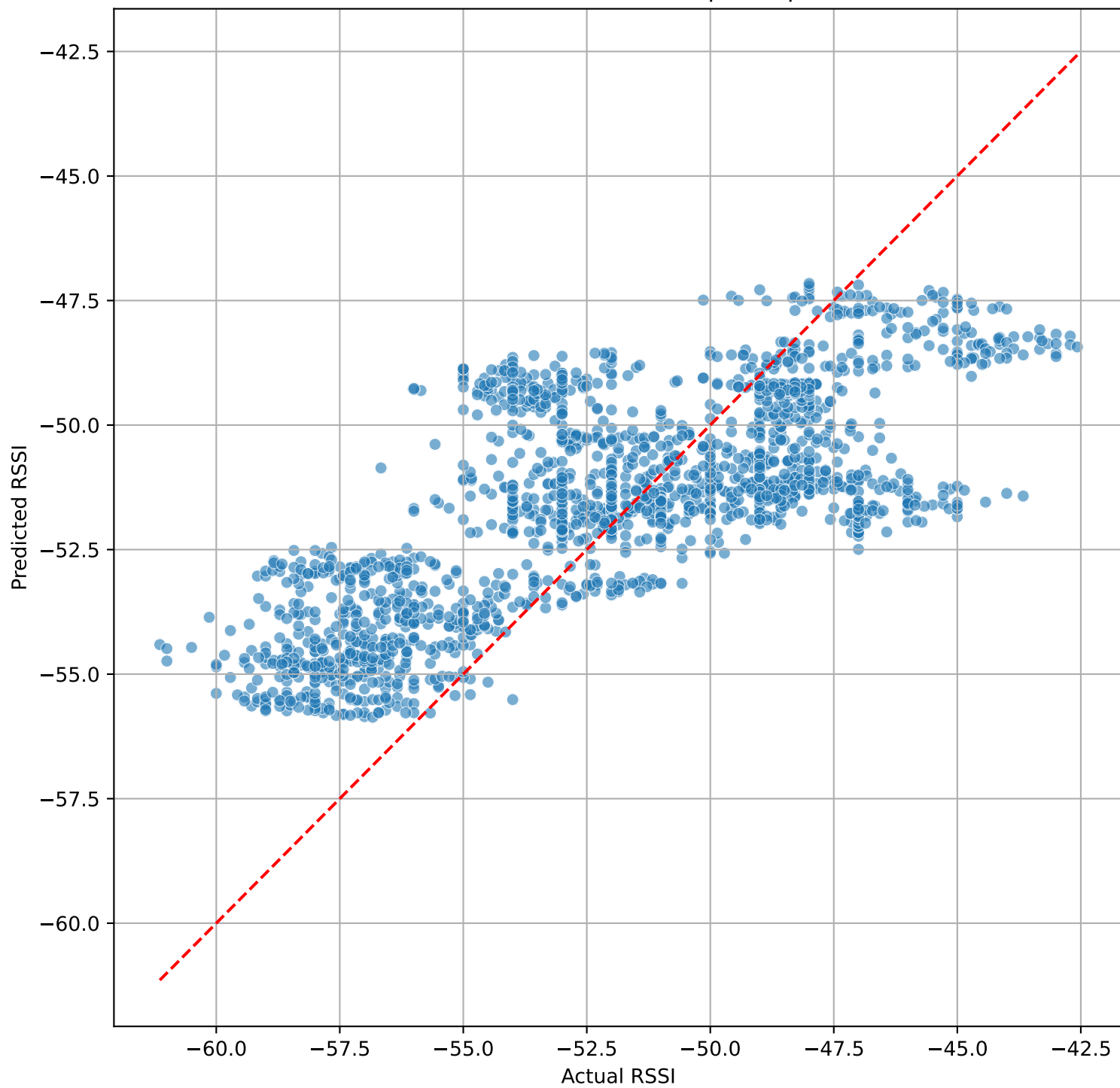
Actual vs Predicted RSSI (Multi-Step (step 48 $\approx$  4.0 s))



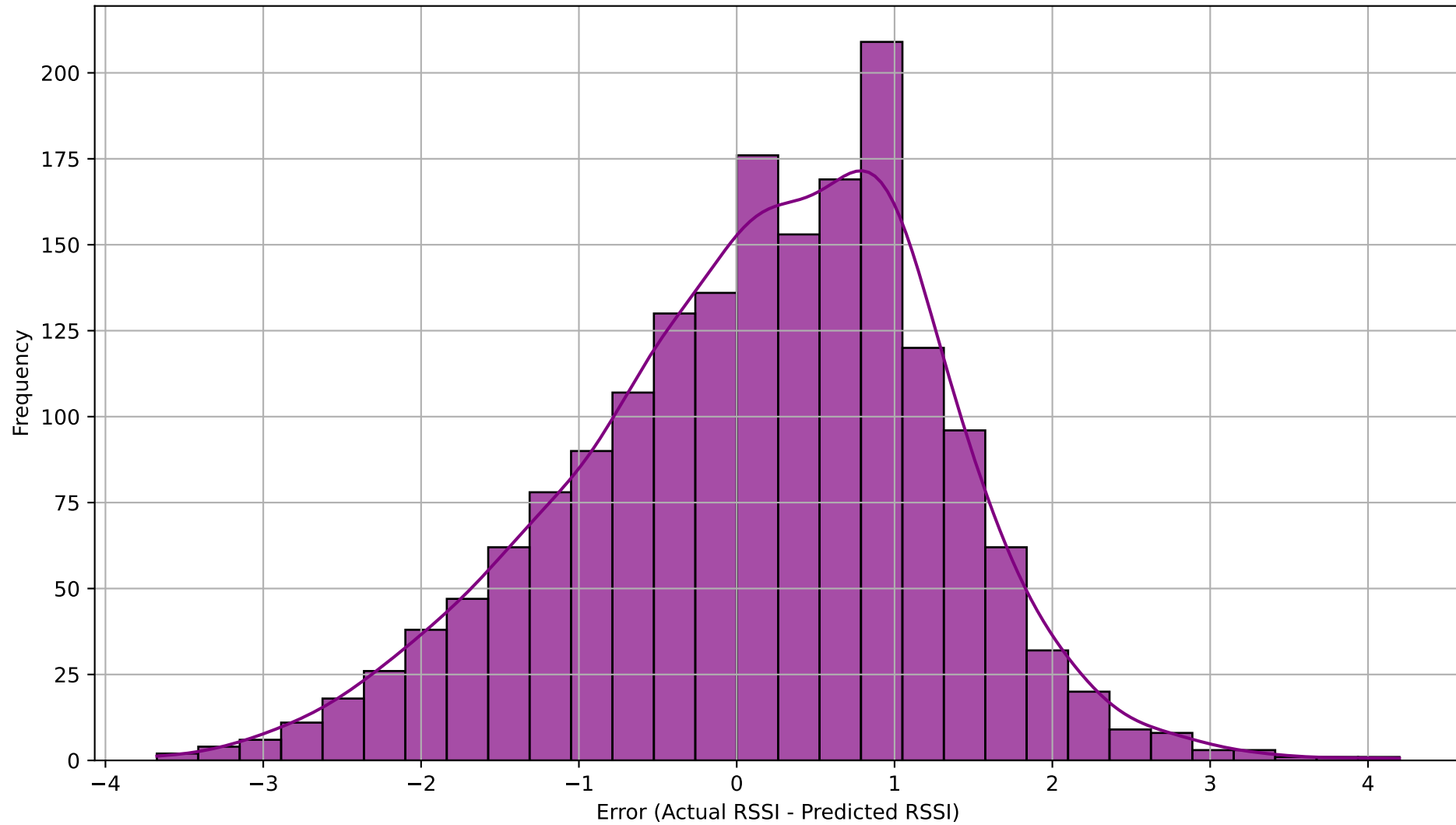
Actual vs Predicted RSSI (First Step)



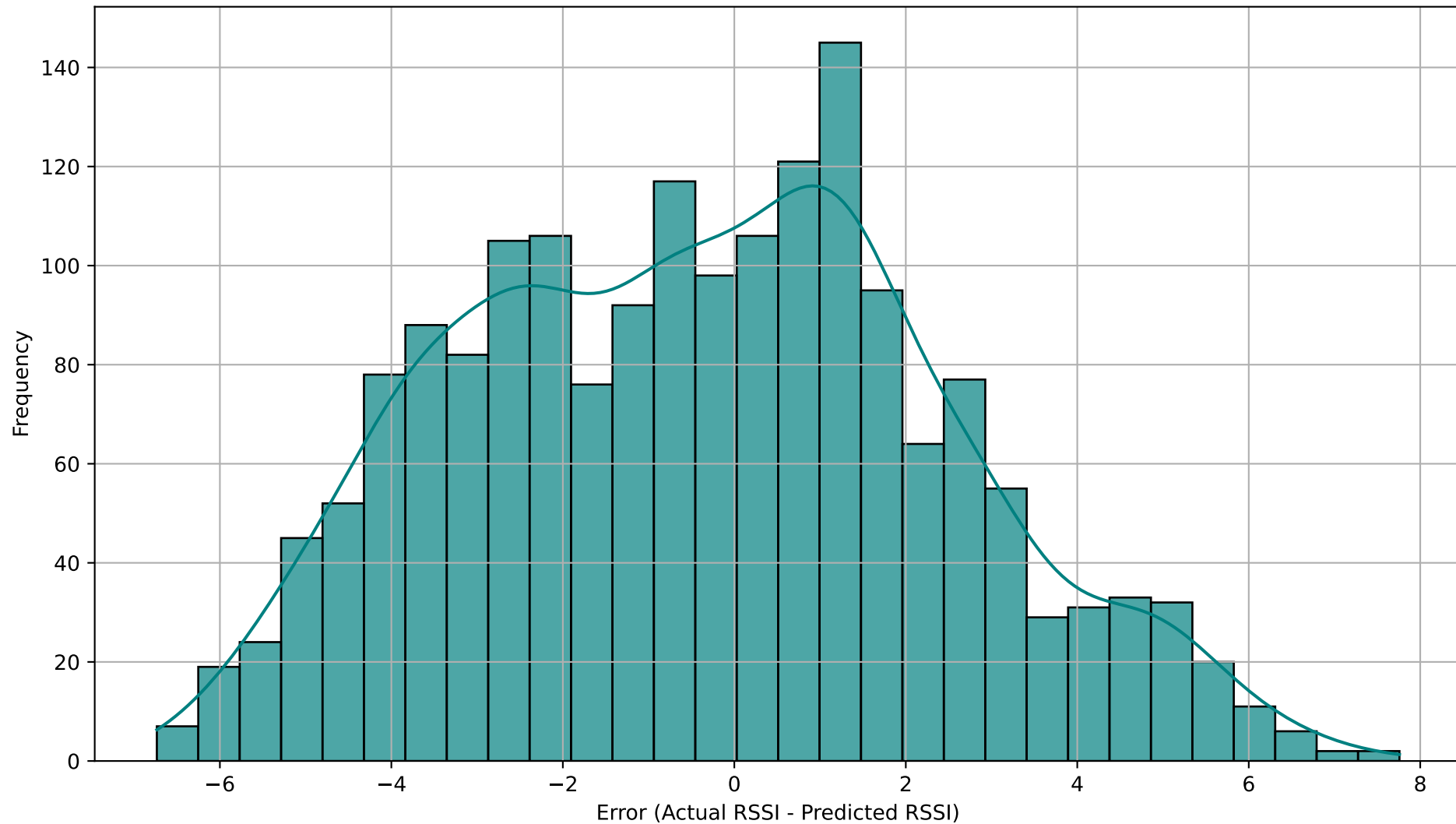
Actual vs Predicted RSSI (Last Step - step 48  $\approx$  4.0 s)



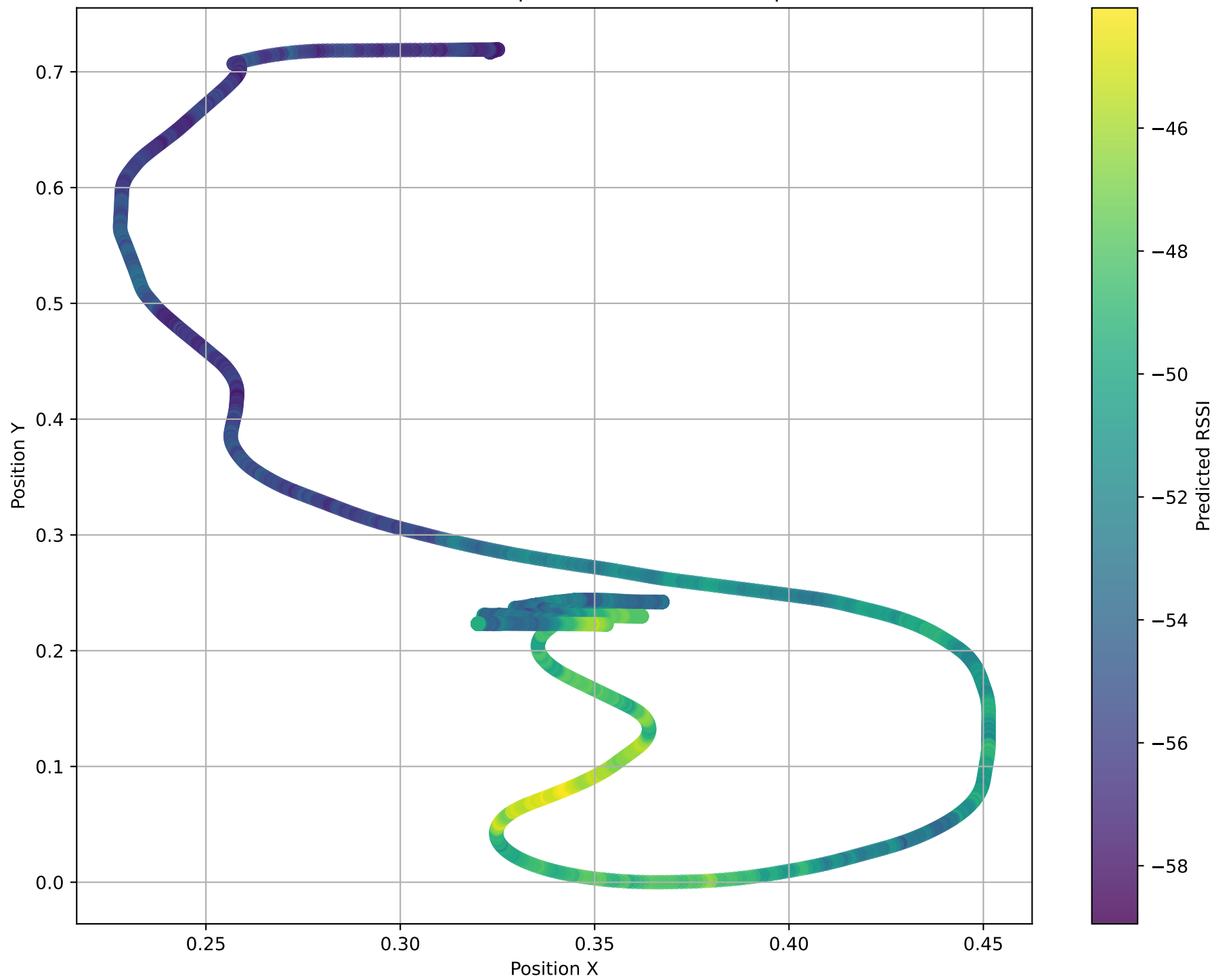
Distribution of Prediction Errors (First Step)



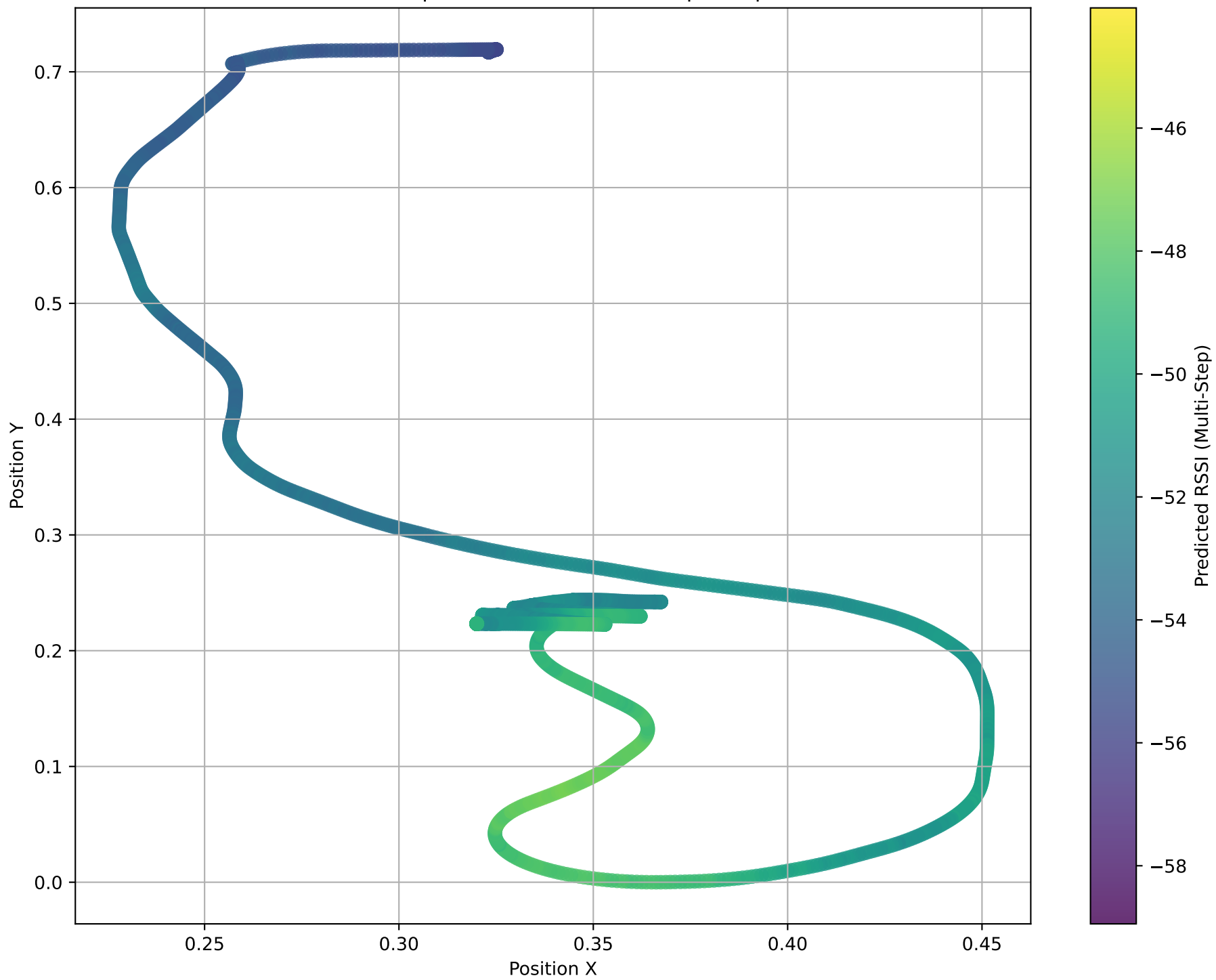
Distribution of Prediction Errors (Last Step - step 48  $\approx$  4.0 s)



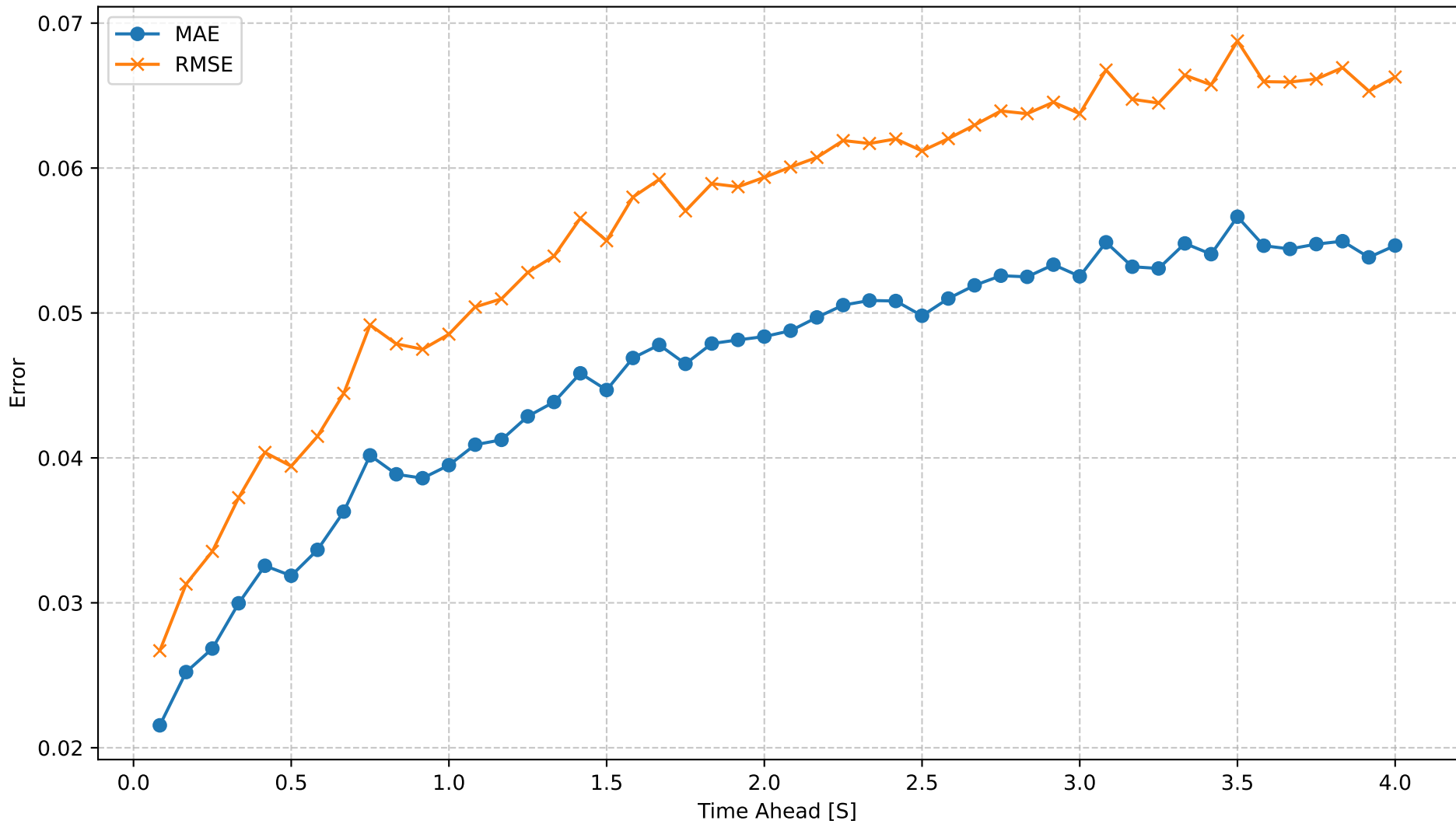
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 48  $\approx$  4.0 s)



Multi-Step Forecast Error by Time Ahead



### Resumo Automático

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Dataset: rssi\_odom\_dataset\_12apr25\_ajust.parquet

Modelo: GRU

Forecast steps: 48

MAE / RMSE (dB)

- 1º passo : 0.02 / 0.03
- Mediana : 0.05 / 0.06
- 48º passo : 0.05 / 0.07

Crescimento do erro: 153.7% → Degrada Rápido (> 50 %)

3 piores passos (MAE): 42 (0.06), 46 (0.05), 37 (0.05)

Recomendação:

Considere aumentar o histórico de entrada ou usar um modelo mais profundo para melhorar multi-step.